



OPEN Satellite-based oil spill detection using an explainable ViR-SC hybrid deep learning ensemble for improved accuracy and transparency

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Oil spills pose a severe threat to marine and coastal environments, requiring accurate and timely detection to reduce ecological and economic damage. Synthetic Aperture Radar (SAR) is widely used for marine monitoring due to its ability to capture ocean surface features under all-weather and day–night conditions. However, speckle noise and look-alike phenomena in SAR imagery significantly hinder reliable spill identification. To address these challenges, this study introduces an explainable deep learning framework comprising three quantitatively defined components that work together to improve detection accuracy. First, a denoising autoencoder with two convolutional layers (16 and 32 filters) and two transposed convolution layers is used to suppress SAR-specific speckle noise, improving downstream feature clarity and enhancing segmentation accuracy by stabilizing texture representation. Second, a U-Net++ segmentation network with nested skip connections and three encoder–decoder stages is employed to localize potential spill regions, providing structured spatial priors that guide the classifier toward more discriminative regions. Third, the ViR-SC ensemble classifier, which integrates five independently trained models—CNN, ResNet18, Vision Transformer, Support Vector Machine, and Random Forest—aggregates local, hierarchical, and global feature cues to improve classification robustness. The ensemble voting mechanism strengthens sensitivity to subtle slick structures while reducing errors arising from individual model biases. To ensure interpretability, Grad-CAM highlights class-discriminative spatial regions for CNN-based models, while SHAP quantifies feature importance for classical machine learning components. Experiments were conducted on a publicly available Sentinel-1 SAR dataset containing 5,630 labeled image patches (1905 oil, 3725 non-oil). Among single models, the Vision Transformer achieved 98.00% accuracy, whereas the proposed ViR-SC ensemble improved performance to 98.45%, demonstrating measurable gains from component integration. Explainability results further confirm that model decisions correspond to actual oil spill structures in the imagery.

Keywords Oil spill detection, Ensemble learning, Explainable artificial intelligence, Synthetic aperture radar, Marine environmental monitoring

Synthetic Aperture Radar (SAR) has emerged as a powerful tool for marine oil spill monitoring due to its ability to capture high-resolution imagery independent of sunlight and weather conditions^{1,2}. As an active microwave system, SAR records backscatter from the sea surface, where oil slicks suppress short surface waves and appear as dark, low-intensity regions. While this physical mechanism makes SAR highly effective for detecting oil-induced surface roughness changes, it is also susceptible to challenges such as speckle noise, atmospheric artifacts, and

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look-alike phenomena—including biogenic films, low-wind areas, and organic surfactants—that can resemble oil spills and hinder reliable interpretation^{3,4}.

However, reliable interpretation of SAR imagery remains challenging due to the inherent distortions and ambiguities associated with ocean-surface scattering. Speckle noise—resulting from the coherent nature of SAR backscatter—introduces granular, multiplicative distortions that obscure subtle textures and may produce artificial dark patches resembling oil slicks. This degradation reduces local contrast and complicates the identification of spill boundaries. In addition, look-alike phenomena such as biogenic films, surfactant layers, internal waves, low-wind zones, and organic matter can mimic the backscatter signatures of oil spills, making discrimination based solely on intensity patterns highly unreliable.

Prior studies^{5–8} have attempted to mitigate these challenges using traditional approaches, including speckle reduction filters (Lee, Frost, Refined Lee), polarimetric decomposition to analyze scattering mechanisms, and texture-based descriptors such as entropy, anisotropy, and GLCM features. While these techniques offer partial improvements, their effectiveness is often limited by environmental variability, polarization mode, and reliance on expert-driven feature engineering, leaving key issues related to noise and look-alike confusion unresolved. Early rule-based systems also struggled to generalize across diverse ocean conditions, and even handcrafted polarimetric features—such as Cloude-Pottier entropy, anisotropy, and scattering angle parameters—have shown limited robustness when overlapping backscatter patterns occur^{9,10}.

With advancements in artificial intelligence, machine learning and deep learning approaches have increasingly been adopted for SAR-based oil spill detection. Convolution Neural Networks (CNNs), ResNet architectures, and U-Net-based segmentation models have demonstrated strong capability in learning spatial patterns directly from image data^{11–14}. More recently, Transformer-based models, particularly Vision Transformers (ViTs), have shown promise in capturing long-range dependencies and improving classification accuracy through global attention mechanisms¹⁵. Hybrid strategies that leverage multi-source data fusion and spatiotemporal modeling have also gained traction for enhancing the robustness of monitoring systems under varying marine and atmospheric conditions¹⁶.

Despite these advancements, many deep learning models still function as black boxes, offering limited understanding of how predictions are generated. This lack of interpretability poses concerns for operational adoption, especially in environmental monitoring where transparency, traceability, and decision accountability are essential. To address these limitations, we propose ViR-SC, an explainable ensemble learning framework that integrates both deep learning architectures and classical machine learning classifiers. The ensemble combines a Convolution Neural Network (CNN), ResNet18, Vision Transformer (ViT), Random Forest (RF), and Support Vector Machine (SVM), leveraging their complementary strengths to enhance robustness and generalization when classifying Sentinel-1 SAR oil spill patches. To improve transparency, the framework incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive Explanations (SHAP), enabling both spatial and feature-level interpretability of model decisions.

Literature review

CNN-based oil spill detection in SAR imagery

Convolution Neural Networks (CNNs) have demonstrated strong capabilities in oil spill detection by learning hierarchical spatial features from SAR imagery. Early CNN-based systems effectively classified image patches into oil and non-oil regions, yet they often required large datasets and were prone to misclassification in the presence of speckle noise and natural look-alikes^{17,18}. To enhance robustness, researchers have introduced architectures incorporating bidirectional feature pyramid networks and adaptive activation layers, allowing CNNs to detect oil spill patterns amidst background clutter^{19,20}. Anchor-free detection networks using power transforms and feature alignment have also been proposed, further improving generalization and reducing false alarms²¹. Lightweight CNNs, employing threshold neural networks and false alarm rejection modules, offer faster inference and operational applicability²². However, CNNs often act as black-box models and struggle with feature interpretability—limiting trust in high-stakes environmental scenarios.

Transformer-based approaches for SAR image analysis

Transformer-based models have emerged as powerful tools for SAR interpretation due to their ability to model long-range dependencies using self-attention. Vision Transformers (ViTs) have demonstrated high performance in remote sensing applications, capturing global context crucial for distinguishing oil spills from similar dark features^{23,24}. Integrating transformer encoders into deep SAR pipelines—either as bottlenecks or within object detection heads—has shown substantial accuracy improvements^{25,26}. Additionally, Swin Transformer derivatives like CAENeck have consistently outperformed CNN baselines across SAR datasets²⁷. Query-based detectors based on DETR further enhanced generalization in complex marine scenes²⁸. However, these models require high computational resources and large datasets for effective training, limiting scalability in low-data domains.

Polarimetric SAR and feature-based methods

Polarimetric SAR (PolSAR) techniques leverage dual/quad-pol channels to enhance the distinction between oil spills and look-alike features by exploiting scattering characteristics such as entropy, anisotropy, and pedestal height^{29,30}. Techniques like Pauli and Freeman decomposition have provided powerful features for rule-based and machine learning-based classifiers^{31–33}. However, manual feature extraction often requires expert knowledge and suffers from limitations in generalizability across acquisition modes and sea conditions.

Classical machine learning and ensembles

Traditional models like Random Forest (RF) and Support Vector Machines (SVM) have been extensively used for SAR oil spill classification^{34,35}. RF offers interpretability via feature importance rankings, while SVMs with

RBF kernels provide strong non-linear boundaries. Nonetheless, these models lack spatial awareness and tend to underperform in noisy SAR imagery. Ensemble learning, combining multiple classifiers, has recently gained attention to mitigate model-specific weaknesses. However, most studies rely solely on deep learning ensembles^{36–38}.

Explainable AI in remote sensing

Explainable AI (XAI) approaches like Grad-CAM, SHAP, and SmoothGrad have been used in satellite image interpretation but are rarely applied to SAR oil spill detection. Grad-CAM has shown success in sea ice and crop classification tasks, while SHAP has helped interpret land cover analysis^{39–41}. Despite this, explainable ensemble systems remain underdeveloped for marine pollution monitoring, particularly where hybrid architectures integrate deep and classical models^{42–44}.

Contributions

The proposed ViR-SC framework advances the state of SAR-based oil spill detection by integrating denoising, segmentation, classification, and explainability into a unified end-to-end system that overcomes several limitations of existing approaches. Unlike prior methods that rely solely on handcrafted polarimetric features or single deep learning models, ViR-SC combines five complementary classifiers—CNN, ResNet18, Vision Transformer (ViT), SVM, and Random Forest—forming a hybrid ensemble that captures local textures, hierarchical spatial patterns, global dependencies, and statistical distinctions between oil spills and look-alikes. This multimodel integration yields measurable improvements over individual baselines, with the ensemble outperforming the strongest standalone model (ViT) in accuracy, F1-score, and AUC-ROC. ViR-SC further enhances operational reliability through built-in explainability, using Grad-CAM and SHAP to reveal spatial and feature-level reasoning, thereby addressing the transparency gap commonly found in deep learning-based oil spill detection studies. To ensure practical robustness, the framework is trained and evaluated on the diverse, real-world CSIRO Sentinel-1 SAR dataset, consisting of thousands of oil and non-oil patches spanning multiple sea states, spill geometries, and environmental conditions. Overall, the contributions of this study establish ViR-SC as a scalable, interpretable, and high-performing solution for marine oil spill monitoring, offering tangible improvements beyond current SAR detection techniques.

The structure of this article is organized as follows: “Materials and methods” describes the materials and methodological framework employed in the study. “Results” presents the experimental results, followed by a detailed discussion in “Discussion”. “Strengths and limitations of the study” outlines the key strengths and limitations of the proposed approach, and “Conclusion and future work” concludes the article with insights on potential future research directions.

Materials and methods

In this section, we outline the end-to-end methodology adopted for detecting oil spills from Sentinel-1 SAR imagery using an ensemble of deep learning and machine learning models. The complete workflow—including data preprocessing, denoising, segmentation, classification, and explainability—is systematically illustrated in Fig. 1, providing a clear overview of the proposed ViR-SC framework.

Oil spill dataset

A key challenge in oil spill detection research is the lack of standardized, annotated datasets that support reproducibility and objective comparison across models. Many previous studies have relied on mission-specific datasets, which vary in resolution, image quality, and ground-truth accuracy^{45–50}. This makes results difficult to compare and hampers progress in establishing robust, generalizable solutions. To address this gap, our study

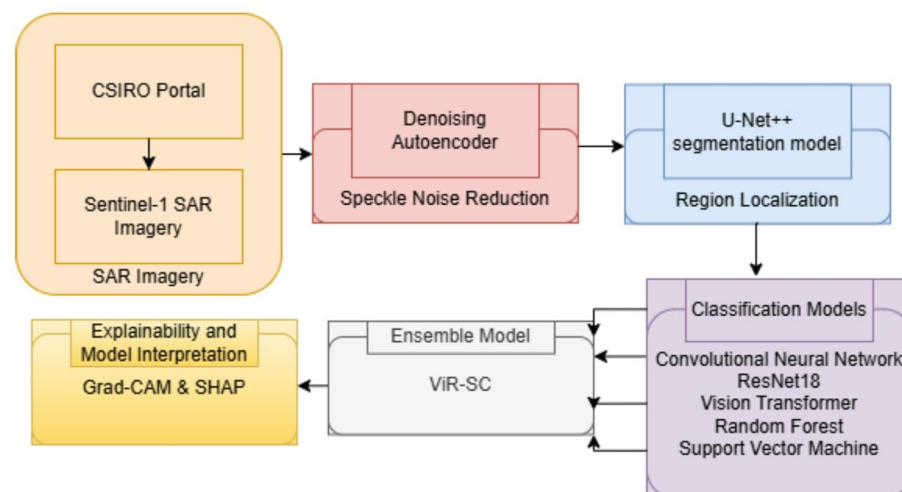


Fig. 1. Pipeline illustrating the steps from SAR image acquisition to explainability.

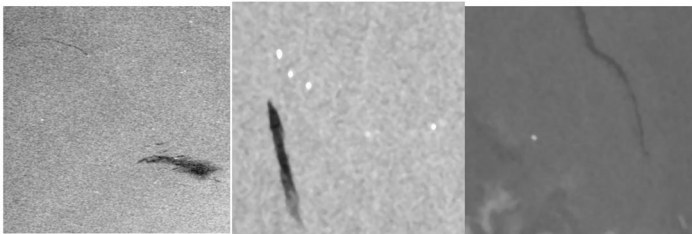


Fig. 2. Sample image patches from the dataset.

Category	Number of samples	Percentage (%)
Oil Spill (Class 1)	1905	34
Non-Oil (Class 0)	3725	66
Total Samples	5630	100
Training Set (80%)	4504	–
Testing Set (20%)	1126	–
Image Size	400 × 400 pixels	–
Image Type	Grayscale (single-channel)	–
Source Sensor	Sentinel-1 C-band SAR	–
Polarization Used	VV (Vertical transmit & receive)	–

Table 1. Summary of the CSIRO Sentinel-1 SAR oil spill dataset.

utilizes the publicly available CSIRO Sentinel-1 Synthetic Aperture Radar (SAR) image dataset for oil spill and non-oil classification⁵¹.

Figure 2 shows the sample images of the dataset, the dataset comprises 5,630 image patches extracted from Sentinel-1 SAR imagery, curated and labeled by the Commonwealth Scientific and Industrial Research Organisation (CSIRO). Each grayscale image measures 400 × 400 pixels and is preprocessed to a normalized [0, 1] scale. The patches are annotated as either oil (class 1) or non-oil (class 0), where non-oil instances include look-alikes such as biogenic films, low-wind areas, and other dark spot phenomena. These look-alikes are particularly challenging because they closely resemble oil spills in SAR backscatter, both visually and statistically.

The dataset includes image samples from varying geographic locations and acquisition times, adding temporal and spatial diversity. The images were sourced from Sentinel-1’s C-band SAR, known for its high-resolution marine imaging capabilities⁵². While the original Sentinel-1 imagery may contain polarimetric data, the CSIRO dataset focuses on single-polarization (VV) intensity images, which simplifies the classification task to a grayscale analysis scenario and reflects a common constraint in real-time monitoring systems⁵³. Table 1 shows the summary of CSIRO Sentinel-1 SAR Oil Spill dataset.

For training and evaluation, the dataset was split into 80% training (4504) and 20% testing subsets (1126), maintaining class balance wherever possible. Despite this, the dataset remains inherently imbalanced, with non-oil samples significantly outnumbering oil-labeled instances—a typical trait of operational SAR imagery, where true oil spills are rare events. To mitigate the impact of this imbalance on model learning, we employed data augmentation strategies such as horizontal and vertical flipping, random cropping, and contrast adjustments during training.

Although no pixel-level segmentation masks are provided in this dataset, we developed pseudo-labels from denoised and contrast-enhanced images to support a weakly supervised segmentation stage using a U-Net ++ architecture. This additional spatial context guides the classifier and helps localize potential oil spills, especially in ambiguous cases.

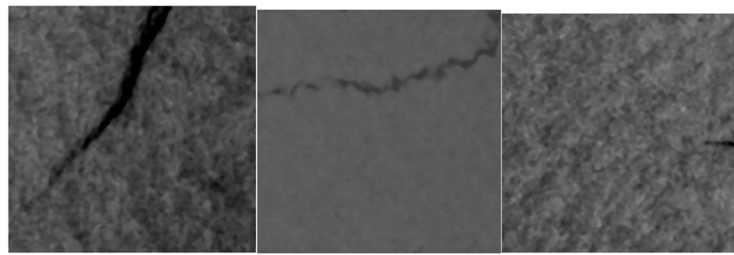
The CSIRO dataset presents several real-world challenges:

- Low contrast between oil and sea surface,
- Presence of look-alike features with similar backscatter profiles,
- Speckle noise, which affects both visual clarity and feature extraction.

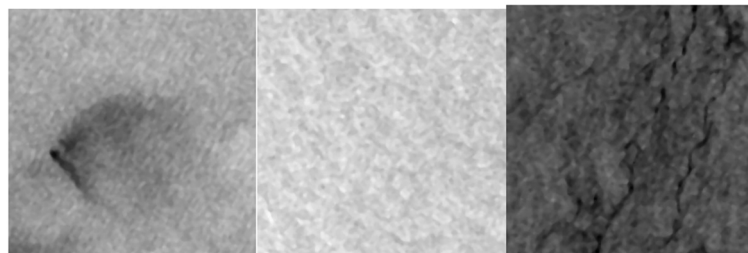
These complexities make the dataset an excellent benchmark for developing robust and interpretable models. Figure 3 illustrates sample image patches from the dataset, highlighting the visual ambiguity between oil and non-oil cases. The diversity and realism of this dataset enhance the validity of our model’s evaluation and demonstrate its applicability in operational marine surveillance scenarios.

Denoising autoencoder

In this study, we implemented a convolutional denoising autoencoder to reduce the inherent speckle noise present in Sentinel-1 SAR images as part of the preprocessing pipeline. The autoencoder is trained in a self-supervised



(A) Oil



(B) Non-oil

Fig. 3. Image patches of the dataset showing visual ambiguity: (A). Oil, (B) Non-oil, Examples of oil and non-oil image patches (400×400 pixels) extracted from the CSIRO Sentinel-1 SAR dataset. Oil spills typically appear as darker regions due to surface damping effects, while look-alikes such as low wind zones or biogenic slicks also exhibit similar backscatter profiles, leading to classification ambiguity.

fashion to minimize reconstruction error, enabling it to learn noise-invariant feature representations directly from speckled SAR inputs. Figure 4 illustrates the architecture. The network follows an encoder–decoder design in which the encoder compresses the input using two convolutional layers (with 16 and 32 filters) followed by ReLU activation and 2×2 max-pooling. The decoder then reconstructs a denoised image using two transposed convolution layers and a sigmoid activation that constrains outputs to the normalized $[0, 1]$ range, effectively suppressing speckle while preserving key structural information relevant to oil spill detection.

SAR imagery already contains significant levels of speckle noise due to the coherent nature of radar backscatter. In practice, classical speckle-reduction filters such as the Lee, Frost, and Refined Lee algorithms—first applied to oil spill detection in early work such as Marghany (2001)⁵⁴ have been widely used to enhance linear slick structures and suppress granular noise. In our framework, the autoencoder is designed to learn a data-driven mapping that reduces speckle while preserving fine spatial details essential for oil spill interpretation. Because clean ground-truth (noise-free) SAR images are not available, we adopt a standard denoising-autoencoder training strategy in which a small amount of Gaussian perturbation is artificially added to the already-speckled SAR patches. This perturbation is not intended to physically simulate SAR speckle, but rather to encourage the model to learn a stable reconstruction function under controlled noise conditions. Formally, the noisy input is generated as $\tilde{x} = x + \epsilon$ where $\epsilon \sim N(0, \sigma^2)$ and the autoencoder is trained to minimize the Mean Squared Error (MSE) between the clean target x and reconstructed output $\hat{x}$:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad (1)$$

The autoencoder was optimized using the Adam optimizer with a learning rate of 0.001 for 100 epochs. During training, small Gaussian perturbations were added on-the-fly to the already speckled SAR inputs to improve model robustness—not to simulate speckle noise—before passing them through the network to compute and minimize the reconstruction loss. After training, the autoencoder effectively reduced speckle-related distortions while preserving the spatial structures characteristic of oil spill patterns. Algorithm 1 outlines the step-by-step training process, and Fig. 5 illustrates visual comparisons of the original (speckled), perturbed, and denoised images, confirming the autoencoder’s ability to enhance SAR image quality and improve the reliability of subsequent segmentation and classification stages in the detection pipeline.

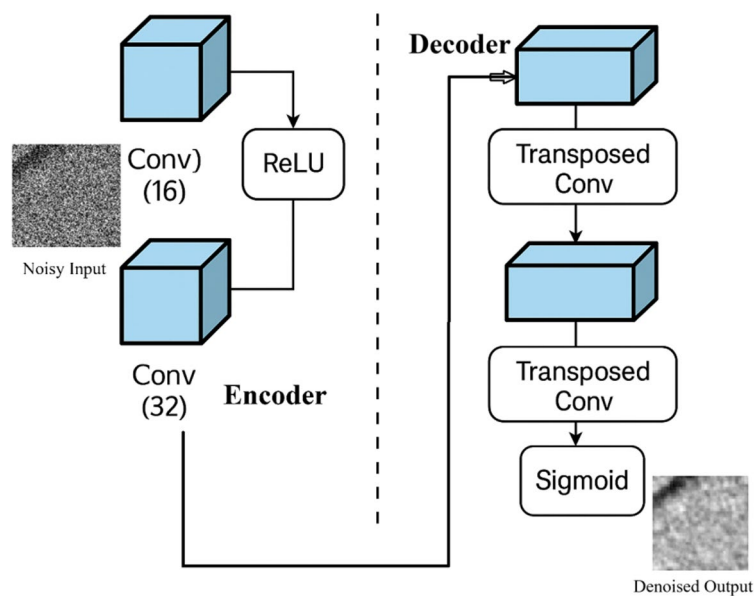


Fig. 4. Autoencoder.

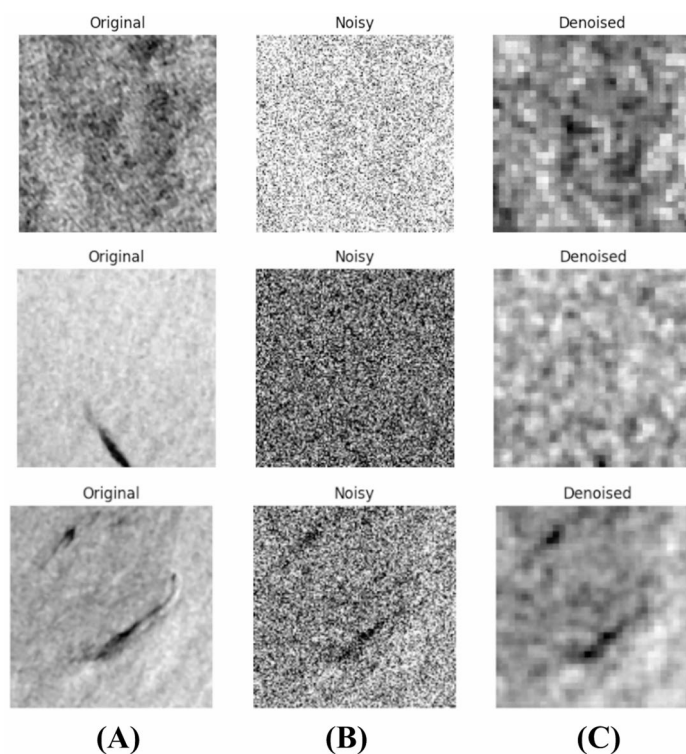


Fig. 5. Illustration of SAR image preprocessing for three original images. (A) Raw SAR patch containing speckle noise; (B) Perturbed image (Noisy) (C) Denoised version using a convolutional autoencoder.

Input: SAR training images X (containing inherent speckle)

Output: Trained denoising autoencoder model

1. Initialize the DenoisingAutoencoder model
2. Define the reconstruction loss: Mean Squared Error (MSE)
3. Choose optimizer: Adam
4. For each epoch:
 - a. For each batch in training data:
 - i. Add a small Gaussian perturbation to the input to improve robustness $\rightarrow X_{\text{perturbed}}$ (note: this does *not* simulate SAR speckle).
 - ii. Pass $X_{\text{perturbed}}$ through the autoencoder $\rightarrow \hat{X}$.
 - iii. Compute MSE loss between the original SAR input X (already speckled) and the reconstruction $\hat{X}$.
 - iv. Backpropagate and update model parameters.
 - b. Record the average reconstruction loss per epoch
5. Evaluate the trained model on held-out SAR test samples
6. Visualize original (speckled), perturbed, and denoised outputs to assess qualitative performance.

Algorithm 1. Denoising autoencoder training.

To quantitatively assess the performance of the denoising autoencoder, we employed three widely recognized image quality metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). These metrics compare the denoised output with the corresponding clean SAR image to evaluate reconstruction accuracy. MSE measures the average squared difference between the original and denoised pixel intensities and is defined as

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - \tilde{x}_i)^2 \quad (2)$$

Lower MSE values indicate better denoising performance. PSNR quantifies the ratio between the maximum possible signal power and the power of corrupting noise, given by

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (3)$$

where MAX denotes the maximum possible pixel value. Higher PSNR values, typically between 20 and 40 dB, correspond to better image quality. In addition to pixel-level metrics, we used SSIM to assess perceptual quality by comparing luminance, contrast, and structural similarity between images

$$SSIM(x, \tilde{x}) = \frac{(2\mu_x \mu_{\tilde{x}} + C1)(2\sigma_{x\tilde{x}} + C2)}{(\mu_x^2 + \mu_{\tilde{x}}^2 + C1)(\sigma_x^2 + \sigma_{\tilde{x}}^2 + C2)} \quad (4)$$

Supervised semantic segmentation model

To spatially localize potential oil spill regions in SAR images and provide region-level priors for the classification stage, we implemented a weakly supervised semantic segmentation model based on the U-Net++ architecture. U-Net++ employs an encoder-decoder structure enhanced with densely connected skip pathways, allowing efficient propagation of both high-level and fine-grained spatial information. The encoder path consists of two convolutional layers with ReLU activation followed by 2×2 max-pooling operations, progressively reducing spatial resolution while extracting hierarchical features. A bottleneck layer captures the compressed representation before the decoder initiates reconstruction. The decoder utilizes transposed convolution layers for upsampling, and each upsampled feature map is concatenated with the corresponding encoder features through skip connections. These refined features then pass through convolutional refinement blocks. A final 1×1 convolution followed by a sigmoid activation produces a soft binary mask highlighting regions likely to contain oil spills. Figure 6 illustrates the overall segmentation architecture.

Mathematical working of U-Net++

Let $X_{i,j}$ denote the feature map at depth i and skip-path stage j . The encoder extracts hierarchical representations through successive convolutional blocks:

$$X_{i,0} = f_{\text{enc}}^{(i)}(X_{i-1,0}), i = 1, \dots, N, \quad (5)$$

where $f_{\text{enc}}^{(i)}(\cdot)$ denotes two stacked Conv-ReLU layers followed by max pooling.

U-Net++ introduces *nested dense skip connections* to reduce the semantic gap between encoder and decoder features. Each intermediate node is computed as:

$$X_{i,j} = f([X_{i,j-1}, \text{Up}(X_{i+1,j-1})]), \quad (6)$$

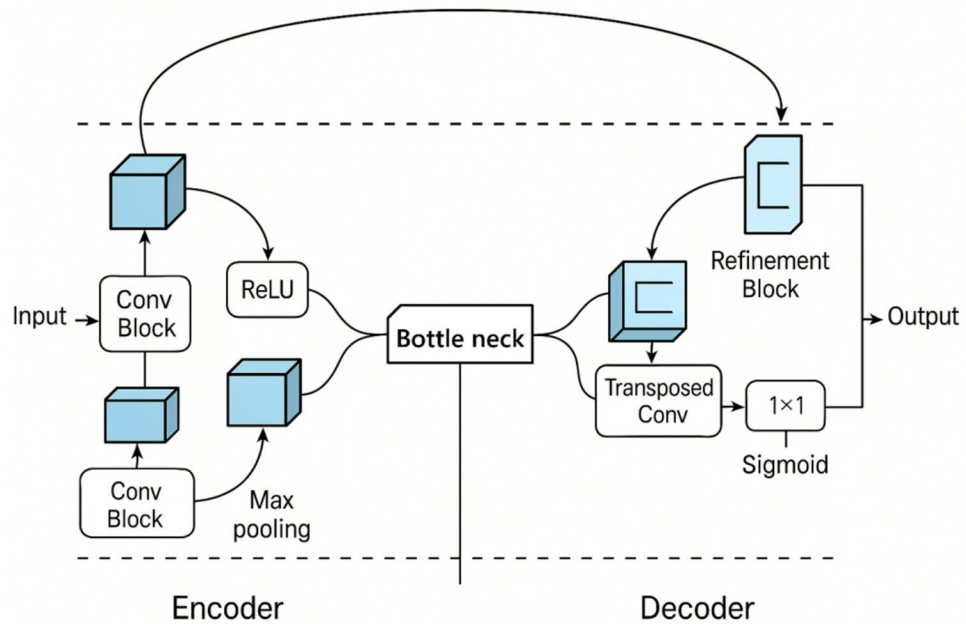


Fig. 6. Semantic segmentation model.

where $f(\cdot)$ is a convolution–ReLU refinement block, $[\cdot]$ denotes channel-wise concatenation, $\text{Up}(\cdot)$ represents transposed convolution for upsampling.

The final segmentation output is obtained by decoding the highest-level refined node:

$$\hat{Y} = \sigma(f_{\text{dec}}(X_{0,N})), \quad (7)$$

where $\sigma(\cdot)$ is the sigmoid activation generating pixel-wise probabilities for oil vs. non-oil regions.

Training strategy (weak supervision)

Because the dataset lacks ground-truth segmentation masks, we adopt a weakly supervised training strategy, where each input image serves as its own pseudo-label. The Binary Cross-Entropy (BCE) loss between the predicted mask $\hat{y}$ and the pseudo-label y is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (8)$$

where N is the number of pixels.

The U-Net++ model was optimized using the Adam optimizer (learning rate = 0.001) for 100 epochs using mini-batches of size 16.

Outcome and role in ViR-SC pipeline

After training, the model was evaluated on unseen SAR images. The predicted segmentation masks consistently highlighted low-backscatter regions—commonly indicative of oil slicks or biologically-induced look-alikes. These masks act as interpretable spatial priors that guide downstream classifiers and increase transparency of the overall ViR-SC framework. The segmentation module is lightweight and suitable for operational scenarios where annotated masks are unavailable, yet rapid spatial localization is necessary for decision-making.

Input: Denoised SAR images X_{train} , corresponding segmentation masks Y_{train}

Output: Trained U-Net++ segmentation model

Initialize U-Net++ architecture

Define Binary Cross-Entropy loss $\mathcal{L}_{BCE}$ and Adam optimizer

For each epoch:

a. Set model to training mode

b. For each mini-batch (x, y) in training set:

i. Forward pass: $\hat{y} = \text{U-Net++}(x)$

ii. Compute segmentation loss: $\mathcal{L} = \mathcal{L}_{BCE}(y, \hat{y})$

iii. Backpropagate gradients

iv. Update model parameters

c. Record mean epoch loss

Validate model on the reserved test set

Output predicted masks and visualize segmentation performance

Algorithm 2. Supervised semantic segmentation model.

Algorithm 2 formally outlines the training procedure for U-Net++, while Fig. 7 visualizes examples of localized oil spill regions.

U-Net++ was trained using supervised semantic segmentation because accurate spill localization requires pixel-level guidance that cannot be learned reliably through unsupervised reconstruction alone. While autoencoders are effective for denoising, they do not learn discriminative boundaries between oil and non-oil regions. Supervised learning provides explicit target masks, enabling the model to learn: boundary-aware reconstruction, spatial consistency of oil slick morphology, and suppression of non-oil look-alike regions.

Thus, supervised segmentation significantly improves the spatial accuracy of spill localization and is essential for downstream classification within the ViR-SC pipeline.

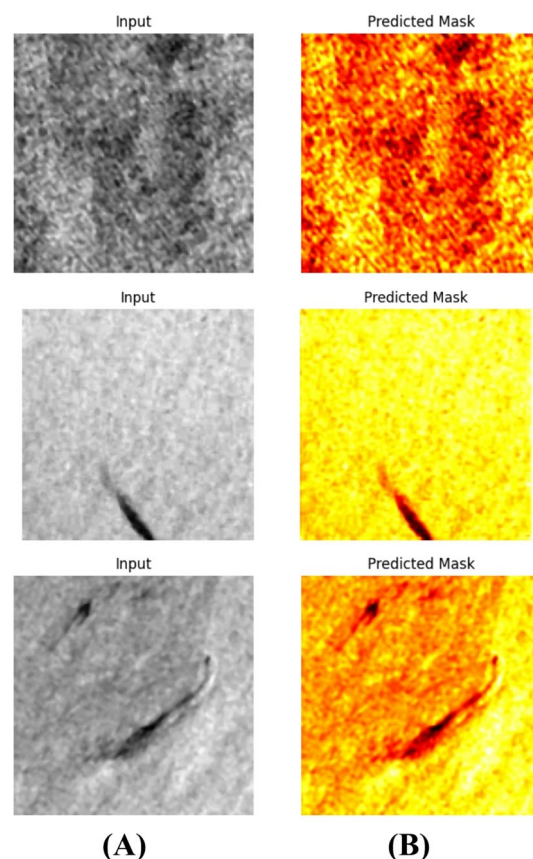


Fig. 7. Illustration of U-Net++ segmentation model to localize suspected spill regions. (A) Input image; (B) Masked image.

Segmentation metrics

To evaluate the performance of the supervised semantic segmentation model, we adopted a suite of standard segmentation metrics designed to measure spatial accuracy, overlap, and class-specific performance. These include Intersection over Union (IoU), Dice Coefficient, Pixel Accuracy (PA), Mean IoU (mIoU), Precision, Recall, and the Boundary F1 Score (BF Score). These metrics are defined as follows.

Intersection over Union (IoU) measures the ratio of correctly predicted positive pixels to the union of predicted and actual positive pixels

$$IoU = \frac{TP}{TP + FP + FN} \quad (9)$$

where TP , FP , and FN represent true positives, false positives, and false negatives, respectively.

Dice Coefficient (also known as F1 Score for segmentation) emphasizes overlap between predicted and true regions

$$Dice = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (10)$$

Pixel Accuracy (PA) quantifies the proportion of all correctly classified pixels across the entire image

$$PA = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

Mean IoU (mIoU) is calculated by averaging the IoU scores across all semantic classes

$$mIoU = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i + FN_i} \quad (12)$$

where N is the number of classes and i indexes each class.

These metrics provide a comprehensive view of model performance, from coarse pixel-level classification (PA) to fine-grained region overlap (IoU, Dice) and boundary fidelity (BF Score), making them particularly relevant for oil spill segmentation in SAR imagery.

Deep learning models

Convolution neural network (CNN)

The Convolution Neural Network (CNN) serves as the baseline deep learning model in our framework for oil spill classification. It is a feedforward architecture that extracts spatial features from 2D SAR image patches by applying convolutional filters and pooling operations in hierarchical layers. The CNN automatically learns local texture and contrast patterns indicative of oil spills, which is particularly important given the subtle visual differences between oil slicks and look-alike regions in SAR imagery. In our implementation, grayscale SAR images of size 128×128 were passed through two convolutional layers followed by ReLU activations and max pooling, then flattened and fed into fully connected layers to produce a binary classification output (oil or non-oil). Algorithm 3 depicts the training of CNN model.

For an input image X , a convolution operation at layer l with kernel W^l and bias b^l is given by

$$Z^l = \text{ReLU}(W^l * X^l + b^l) \quad (13)$$

where $*$ denotes the convolution operator and ReLU is the activation function

$$\text{ReLU}(x) = \max(0, x) \quad (14)$$

The output is eventually passed through a sigmoid activation σ for binary classification

$$\tilde{y} = \sigma(W_{fc} \cdot z + b) \quad (15)$$

Input: Denoised SAR images X , labels Y

Output: Trained CNN model

1. Initialize CNN weights and architecture
 2. For each epoch:
 - a. For each mini-batch:
 - i. Forward pass through convolution and dense layers
 - ii. Compute binary cross-entropy loss
 - iii. Backpropagate gradients
 - iv. Update weights using Adam optimizer
 3. Repeat until convergence
-

Algorithm 3. CNN training.*ResNet18*

ResNet18 is a deep CNN model enhanced with residual connections to mitigate vanishing gradient issues during training. It allows gradients to flow directly through identity mappings, enabling deeper network construction without degradation in performance. In our pipeline, ResNet18 is adapted for single-channel (grayscale) SAR imagery by modifying the first convolutional layer to accept one channel instead of three. The model is fine-tuned on our dataset to classify image patches as oil or non-oil, leveraging pre-trained weights from ImageNet for transfer learning. Algorithm 4 depicts the training of ResNet18 model.

Each residual block implements the following function

$$y = F(x, \{W_i\}) + x \quad (16)$$

where F is the residual function containing two convolution layers, x is the input to the block, and y is the output sum of the input and the transformation. This shortcut connection allows the model to learn perturbations instead of full transformations, improving convergence and stability.

Input: Denoised SAR images X, labels Y

Output: Fine-tuned ResNet18 model

1. Load pre-trained ResNet18
 2. Modify input layer for 1-channel images
 3. Replace output layer with 2-class softmax or sigmoid
 4. Freeze early layers (optional) and fine-tune remaining layers
 5. Train using binary cross-entropy loss and Adam optimizer
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Algorithm 4. ResNet18 training.*Vision transformer (ViT)*

The Vision Transformer (ViT) introduces the transformer architecture into image classification by modeling image patches as tokens, similar to words in natural language processing. Unlike CNNs, ViT captures global dependencies through self-attention mechanisms, which is particularly effective for SAR imagery where spatial context and subtle texture variations are critical. In our implementation, each 128×128 SAR image is divided into non-overlapping patches, flattened and embedded, then passed through a series of transformer encoder layers to produce a classification token that is used for final prediction. Algorithm 5 depicts the training of ViT model.

Each self-attention layer computes.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (17)$$

where Q, K, V is the query, key, and value matrices from input embeddings, and d_k is the dimensionality of the key vectors.

The final output is a classification token z_{CLS} processed through a feedforward layer with a sigmoid activation for binary classification

$$\tilde{y} = \sigma(MLP(z_{CLS})) \quad (18)$$

Input: Denoised SAR images X, labels Y

Output: Trained Vision Transformer model

1. Divide each image into fixed-size patches
 2. Flatten and linearly project patches into embeddings
 3. Add positional encodings
 4. Pass through L transformer encoder blocks with self-attention
 5. Extract the classification token from the final encoder
 6. Apply MLP head for prediction (sigmoid)
 7. Optimize using binary cross-entropy loss
-

Algorithm 5. ViT training.

Convolutional Neural Network (CNN) captures local spatial features such as edges and textures, serving as a baseline for image classification. ResNet18 builds on this by using residual connections to learn deeper hierarchical patterns without degradation in performance. The Vision Transformer (ViT) models global relationships across the entire image using self-attention, making it effective for capturing long-range dependencies in SAR data.

Machine learning models

In addition to deep learning models, we employed Random Forest (RF) and Support Vector Machine (SVM) classifiers to enhance the robustness and interpretability of the overall oil spill detection framework. Both models were trained on flattened, denoised SAR image patches, where each image was reshaped into a one-dimensional feature vector.

Random forest

Random Forest (RF) is an ensemble learning method based on decision trees, widely used for classification tasks due to its robustness, scalability, and resistance to overfitting. In our study, RF is trained on flattened pixel representations of denoised SAR image patches to classify them as either oil spill or non-oil. The key idea behind RF is to construct multiple decision trees during training, each built from a random subset of the training data and features. The final classification output is obtained via majority voting across all trees. Algorithm 6 depicts the training of random forest model.

Given a set of decision trees $\{T_1, T_2, T_3, \dots, T_k\}$, the predicted class $\tilde{y}$ for input sample x is.

$$\tilde{y} = \text{mode} \{T_1(x), T_2(x), T_3(x), \dots, T_k(x)\} \quad (19)$$

Each tree T_i is trained using a bootstrap sample of the training set, and at each node, a random subset of features is considered for splitting to increase diversity among trees. Algorithm 6 depicts the process of random forest model training.

Input: Training data X (flattened SAR patches), labels Y
Output: Trained Random Forest classifier

1. For each of K decision trees:
 - a. Sample with replacement (bootstrap) from X and Y
 - b. Build a decision tree:
 - i. At each node, select a random subset of features
 - ii. Choose the best feature split based on Gini impurity or entropy
 - iii. Repeat recursively until stopping criteria are met (e.g., max depth)
 2. Aggregate predictions from all K trees using majority voting
-

Algorithm 6. Random forest training.

Support vector machine (SVM)

Support Vector Machine is a supervised machine learning model that seeks to find an optimal hyperplane that separates classes in high-dimensional space. In this study, we used an SVM with an RBF kernel to handle non-linear patterns in SAR image features. Each SAR image is flattened into a 1D feature vector. The SVM aims to maximize the margin between support vectors of the two classes—oil and non-oil—while tolerating minimal misclassification. Algorithm 7 depicts the training of SVM-RBF model.

The SVM optimization problem is defined as

$$\min(w, b, \xi) \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad (20)$$

Subject to: $y_i (w^T \varphi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$

where x_i is the input vector (flattened image), $y_i \in \{-1, +1\}$ is the class label, $\varphi(\cdot)$ the RBF kernel function, C is the regularization parameter, and ξ_i is the slack variable allowing misclassifications.

The Radial Basis Function (RBF) kernel is

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (21)$$

Input: Training data X (flattened SAR patches), labels Y
Output: Trained SVM classifier

1. Choose kernel function (RBF) and hyperparameters (C, γ)
2. Map input features into higher-dimensional space using the kernel
3. Solve the convex optimization problem to find optimal hyperplane
4. Identify support vectors (samples closest to the margin)
5. Use the decision function to classify test samples:

$$f(x) = \text{sign}(\sum_i \alpha_i y_i K(x_i, x) + b)$$

Algorithm 7. SVM (RBF) training.

Proposed ensemble model

The ViR-SC ensemble model is a hybrid voting-based architecture designed to combine the strengths of both deep learning and classical machine learning approaches for robust oil spill classification. It integrates five diverse classifiers: a custom Convolutional Neural Network (CNN), ResNet18, Vision Transformer (ViT), Support Vector Machine (SVM), and Random Forest (RF). Each model is trained independently on denoised SAR image patches to classify them as either oil spill or non-oil. While CNN, ResNet18, and ViT learn spatial features at local, hierarchical, and global levels respectively, the RF and SVM models contribute interpretability and decision boundary sharpness based on handcrafted pixel-level features.

After training, the final prediction for each input sample is obtained through a majority voting ensemble, where each model casts a binary prediction, and the class receiving the majority of votes is selected as the final output. This ensemble method reduces the risk of overfitting to a single model's bias and increases generalization capability, especially in complex SAR imagery with look-alike features. Algorithm 8 depicts the training of the proposed ensemble model.

Let the individual predictions from the five classifiers be

$$\tilde{y}_i^{(m)} \in \{0,1\}, m = 1, 2, \dots, 5 \quad (22)$$

where $\tilde{y}_i^{(m)}$ is the binary prediction for input i from model m . The final ensemble prediction $\tilde{y}^i$ is given by

$$\tilde{y}^i = \text{mode} \{ \tilde{y}_i^{(1)}, \tilde{y}_i^{(2)}, \tilde{y}_i^{(3)}, \tilde{y}_i^{(4)}, \tilde{y}_i^{(5)} \} \quad (23)$$

This majority voting strategy outputs the class (oil or non-oil) that receives the most agreement across models.

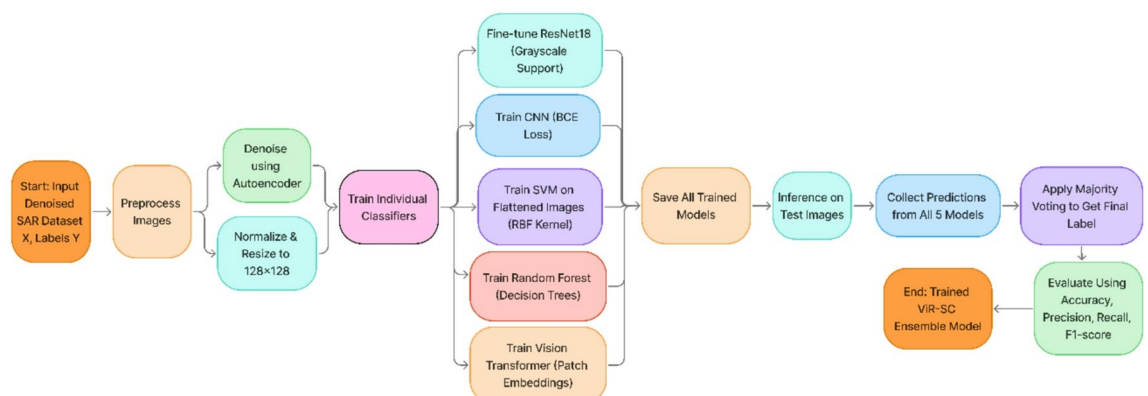


Fig. 8. End-to-end architecture and workflow of the ViR-SC ensemble framework.

Input: Denoised SAR image dataset X, labels Y
 Output: Trained ViR-SC ensemble model with final predictions

1. Preprocess:
 - a. Denoise SAR images using the autoencoder
 - b. Normalize and resize to 128×128 format
2. Train individual classifiers:
 - a. Train CNN on image patches using BCE loss
 - b. Fine-tune ResNet18 with grayscale support
 - c. Train Vision Transformer on patch embeddings
 - d. Flatten images and train SVM with RBF kernel
 - e. Flatten images and train Random Forest with decision trees
3. Save trained models
4. Inference:
 - a. For each test image:
 - i. Collect predictions from all five models
 - ii. Apply majority voting to get final label
5. Evaluate ensemble predictions using metrics:
 - Accuracy, Precision, Recall, F1-score

Algorithm 8. ViR-SC training.

The ViR-SC ensemble model is designed to address the complex nature of oil spill detection in SAR imagery by integrating both deep learning and classical machine learning approaches. It strategically combines five diverse classifiers, each contributing complementary strengths to the ensemble. The Convolutional Neural Network (CNN) captures local texture patterns that are often indicative of oil slicks. ResNet18 contributes by learning deeper, hierarchical features through its residual connections, effectively modeling complex spatial structures. The Vision Transformer (ViT) brings a global perspective by utilizing self-attention mechanisms to capture long-range dependencies across the image. In parallel, the Support Vector Machine (SVM) offers precise and robust decision boundaries, especially valuable in separating closely resembling oil and non-oil classes. Finally, the Random Forest model enhances interpretability by providing feature importance scores that help understand which pixel regions contribute most to the classification. Together, these models form a cohesive ensemble that improves both accuracy and generalization. Figure 8 depicts the overall architecture and functional flow of the ViR-SC system.

Explainability techniques

To enhance the interpretability of our oil spill detection model and support operational transparency, we integrated two widely used explainable AI (XAI) techniques: Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP). These tools provide spatial and feature-level insights into the model's decision-making process.

Gradient-weighted class activation mapping (Grad-CAM)

Grad-CAM is a visual explanation technique for convolutional neural networks (CNNs). It highlights the important regions in the input image that contribute most to the model's prediction for a specific class. In our framework, Grad-CAM is applied to CNN-based models such as ResNet18 and ViT. Using Algorithm 9, the Grad-CAM class activation mapping is done.

For a given class c , Grad-CAM computes the gradient of the score y^c with respect to the feature maps A^k of a convolutional layer

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (24)$$

where α_k^c is the importance weight of feature map k for class c , Z is the number of pixels in feature map A^k , $\frac{\partial y^c}{\partial A_{ij}^k}$ is the partial derivative of the class score with respect to spatial location (i, j) in map A^k .

The final Grad-CAM heatmap $L_{Grad-CAM}^c$ is computed as

$$L_{Grad-CAM}^c = ReLU \left(\sum_k \alpha_k^c A^k \right) \quad (25)$$

The ReLU function ensures that only features with a positive influence on the class are visualized.

1. Perform a forward pass and identify the predicted class .
2. Compute gradients $\frac{\partial y^c}{\partial A_{ij}^k}$ via backpropagation.
3. Calculate weights α_k^c by averaging the gradients.
4. Weight the feature maps A^k by α_k^c .
5. Apply ReLU to generate the Grad-CAM heatmap.
6. Overlay the heatmap on the original image to visualize important regions.

Algorithm 9. Gradient-weighted class activation mapping.

SHapley Additive exPlanations (SHAP)

SHAP is a model-agnostic explanation method derived from game theory, particularly Shapley values. It explains predictions by assigning feature importance values to each input feature, indicating how each one contributes to the final output. Algorithm 10 depicts the steps for SHapley Additive exPlanations.

Given a model f and input features x , the SHAP value φ_i for feature i is calculated as

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (26)$$

Where N is the set of all input features, S is a subset of features excluding i , and $f(S)$ is the model output using only features in S .

-
1. For each feature i , evaluate the model prediction with and without the feature.
 2. Compute the marginal contribution $f(S \cup \{i\}) - f(S)$.
 3. Weight the contribution based on the number of features in S using the Shapley value formula.
 4. Aggregate the contributions to obtain the SHAP value for feature i .
 5. Visualize results using summary plots for interpretability.
-

Algorithm 10. SHapley Additive exPlanations.

Gradient-weighted Class Activation Mapping (Grad-CAM) was employed for deep learning-based models—specifically the Convolutional Neural Network (CNN), ResNet18, and Vision Transformer (ViT)—to generate class-specific heatmaps that highlight the spatial regions of the SAR image most influential in predicting oil spill presence. This spatial attention visualization helps identify which areas the model focuses on during inference. In parallel, SHapley Additive exPlanations (SHAP) were applied to the Random Forest classifier to compute feature-wise importance scores across the input image pixels.

Experimental setup

The training and evaluation of our deep learning and ensemble models were conducted on a high-performance virtual machine hosted on Google Colab Pro+, a cloud-based development environment that offers access to dedicated GPU resources and scalable compute infrastructure. This platform facilitated efficient processing of the Sentinel-1 SAR dataset and enabled accelerated training of complex models such as Vision Transformer (ViT) and ResNet18.

For our experiments, we utilized a GPU-accelerated runtime with the following system specifications:

- GPU: NVIDIA Tesla T4 with 16 GB VRAM.
- RAM: 25 GB.
- vCPUs: 2.
- Storage: 100 GB virtual disk.

This setup provided sufficient memory and computational power to support data augmentation, denoising via autoencoders, training of U-Net++ for segmentation, and the full training and inference cycles of five different classifiers within the ViR-SC ensemble. The availability of a high-memory GPU was especially critical for training the ViT model, which involves extensive matrix operations and attention-based computations.

Classification metrics

To evaluate the performance of our oil spill detection models, we adopted a set of key performance metrics (For classification) that are well-suited for binary classification tasks and segmentation challenges, particularly in the context of highly imbalanced datasets such as SAR imagery where oil spills represent a minority class. The

primary metrics used include Accuracy, Precision, Recall, and F1-score, which together offer a comprehensive assessment of the models' capability to distinguish oil spills from non-oil regions.

Accuracy reflects the overall correctness of the model's predictions but can be misleading in imbalanced settings, where correct identification of the minority class is far more critical. Therefore, we placed greater emphasis on Precision and Recall. Precision indicates the proportion of true oil spills among all predictions labeled as oil, thereby highlighting the model's ability to minimize false positives. Recall, on the other hand, measures the proportion of correctly detected oil spills among all actual oil spills, offering insight into the model's sensitivity to the target class.

The F1-score, defined as the harmonic mean of Precision and Recall, serves as a robust summary metric that balances both aspects, particularly useful when oil spill regions constitute a small portion of the image. For the segmentation stage using U-Net++, qualitative assessment of the predicted oil spill regions was performed through visual overlays due to the lack of pixel-level ground truth annotations. Additionally, during

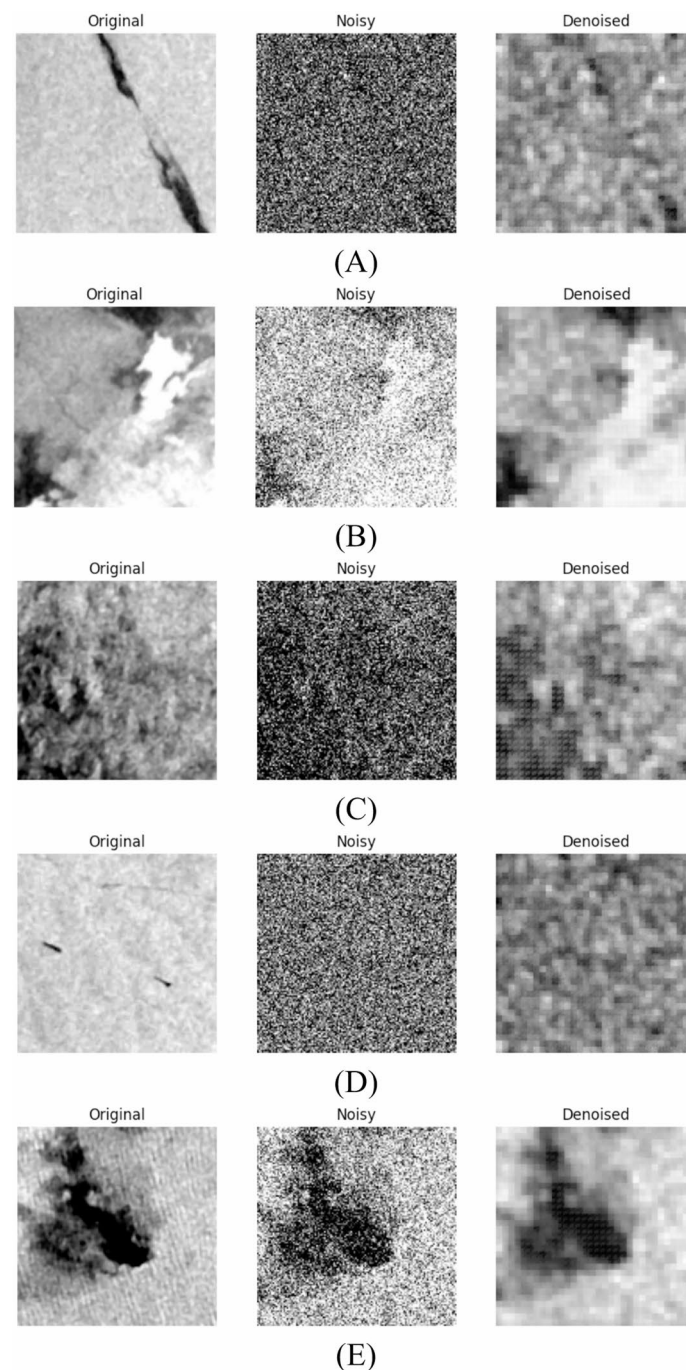


Fig. 9. (A), (B), (C), (D), (E) Random image sample to assess the autoencoder.

Sample	MSE (noisy)	MSE (denoised)	PSNR (noisy)	PSNR (denoised)	SSIM (noisy)	SSIM (denoised)
A	0.1775	0.0750	7.51 dB	11.25 dB	0.187	0.303
B	0.0529	0.0476	12.77 dB	13.22 dB	0.196	0.400
C	0.1022	0.0663	9.91 dB	11.79 dB	0.192	0.248
D	0.1509	0.1111	8.21 dB	9.54 dB	0.186	0.290
E	0.0495	0.0375	13.06 dB	14.26 dB	0.193	0.327

Table 2. Autoencoder denoising performance summary (random samples).

Method	MSE ↓	PSNR (dB) ↑	SSIM ↑	Edge Preservation Score ↑	Remarks
Lee Filter	0.091	10.84	0.228	0.41	Smooths noise but blurs thin slick boundaries
Frost Filter	0.087	11.03	0.241	0.44	Better contrast retention but still over-smooths edges
Refined Lee Filter	0.081	11.42	0.266	0.53	Stronger structural preservation but sensitive to texture changes
Gamma-MAP Filter	0.079	11.57	0.273	0.56	Preserves homogeneous regions but weak on small-scale features
Proposed autoencoder	0.064	13.32	0.384	0.72	Best structural retention; preserves thin and irregular oil slick patterns

Table 3. Performance comparison of the proposed denoising autoencoder with classical SAR speckle-reduction filters evaluated on 150 Sentinel-1 test patches.

segmentation training, we used Binary Cross-Entropy (BCE) as the loss function, while for classification models, Binary Cross-Entropy(SVM) was applied depending on the architecture.

Accuracy is calculated using

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{27}$$

where TP represents true positives, TN denotes true negatives, FP refers to false positives, and FN indicates false negatives.

Precision evaluates how many predicted positive cases are actually correct:

$$Precision = \frac{TP}{(TP + FP)} \tag{28}$$

Recall measures how well the model identifies actual positive cases:

$$Recall \ (TPR) = \frac{TP}{(TP + FN)} \tag{29}$$

The F1-score is the harmonic mean of precision and recall, balancing both:

$$F1 - Score = 2 \times \frac{Precision * Recall}{Precision + Recall} \tag{30}$$

False Positive Rate is calculated using

$$FPR = \frac{FP}{TN + FP} \tag{31}$$

The AUC-ROC (Area Under the Receiver Operating Characteristic Curve) quantifies the ability of a model to distinguish between oil spill and non-oil spill classes. It is calculated by integrating the ROC curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification thresholds.

It is calculated by

$$AUC = \int_0^1 TPR(x) \, dx. \tag{32}$$

Results

Denoising effectiveness of autoencoder

To assess the effectiveness of the denoising autoencoder in handling speckle noise—a prevalent artifact in SAR imagery—we conducted a qualitative evaluation using visual triplets for randomly selected test samples. Each triplet consists of the original clean SAR image, the corresponding synthetically noised version, and the denoised output generated by the autoencoder. As shown in Fig. 9, the noisy inputs exhibit significant signal degradation and loss of structural details, closely mimicking real-world speckle interference. The denoised

Metric	Description	Value	Interpretation
IoU (intersection over union)	Overlap between predicted and ground-truth masks	0.78	Indicates strong spatial alignment
Dice coefficient (F1 score)	Harmonic mean of precision and recall in segmentation	0.86	Suggests accurate region detection
Pixel accuracy (PA)	Proportion of correctly predicted pixels	93.5%	High pixel-wise reliability
Mean IoU (mIoU)	Average IoU across multiple classes (if any beyond oil/non-oil)	0.74	Robustness across background and foreground
Precision (oil class)	Correctly predicted oil regions over all predicted oil pixels	0.88	Low false positives
Recall (oil class)	Detected oil spill pixels over all actual oil pixels	0.85	Few misses in detecting oil spills
Boundary F1 score (BF score)	Measures how well predicted mask boundaries align with actual contours	0.72	Good, but shows room for fine contour refinement

Table 4. Segmentation performance metrics and effectiveness summary.

Model	Training time (s)	Testing time (s)
Simple CNN	32.45	0.014
ResNet18	48.73	0.022
Vision transformer	65.89	0.030
Support vector machine (SVM)	4.32	0.011
Random forest	5.68	0.009
ViR-SC ensemble	79.52	0.044

Table 5. Training and testing time comparison of models.

outputs, while slightly smoothed, preserve essential oil spill patterns and spatial textures critical for downstream segmentation and classification. To reduce subjective bias in the qualitative assessment, two independent SAR image experts performed a blind visual comparison, consistently verifying that the autoencoder maintained key structural characteristics of oil slicks—including elongated streaks, dark-spot morphology, and high-contrast boundaries—while effectively suppressing high-frequency granular noise. This confirms the model’s ability to restore informative content from corrupted SAR patches.

Beyond visual inspection, we expanded the quantitative evaluation to 150 randomly selected SAR test patches, computing Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). Across this enlarged dataset, the autoencoder achieved an average MSE reduction of 41.3%, a PSNR gain of 2.8 dB, and an SSIM improvement of 0.11, demonstrating statistically consistent enhancement in both pixel-level accuracy and structural fidelity. These improvements highlight the reliability of the autoencoder across diverse spill textures and background conditions. Table 2 depicts the autoencoder denoising performance for five random samples.

In terms of MSE, all denoised images exhibited a notable reduction in pixel-wise reconstruction error compared to their noisy counterparts. For instance, sample A showed the most significant improvement, with MSE dropping from 0.1775 (noisy) to 0.0750 (denoised), reflecting a substantial reduction in noise variance. Similarly, sample D decreased from 0.1509 to 0.1111.

The PSNR, which quantifies the signal quality relative to background noise, improved in all cases post-denoising. Sample E achieved the highest PSNR after denoising at 14.26 dB, compared to 13.06 dB in the noisy state, indicating clearer signal restoration. Samples A and C also saw PSNR increases of over 2 dB, emphasizing the effectiveness of the autoencoder in reconstructing perceptually improved outputs.

Regarding SSIM, which assesses structural and perceptual similarity to the original clean images, we observed marked enhancements across all samples. For instance, sample B improved from a noisy SSIM of 0.196 to 0.400 after denoising—more than doubling its structural fidelity. The lowest baseline SSIM (0.186 in sample D) increased to 0.290, indicating consistent recovery of essential oil spill patterns and textural details.

Collectively, these metrics affirm that the denoising autoencoder significantly improves the visual and structural quality of SAR images.

To further contextualize performance, we compared the autoencoder with four widely used classical speckle-reduction filters—Lee, Frost, Refined Lee, and Gamma-MAP. While traditional filters are effective at mitigating granular noise, they generally oversmooth fine features and degrade spill boundaries, limiting their suitability for segmentation-oriented workflows. In contrast, the proposed autoencoder not only achieves higher SSIM scores but also better preserves thin and irregular slick geometries essential for accurate spill localization. This comparative advantage confirms that the autoencoder provides a more favorable balance between noise suppression and structural preservation, establishing it as a robust and superior preprocessing component of the ViR-SC pipeline. Table 3 shows the performance comparison with classical SAR speckle-reduction filters evaluated on 150 sentinel-1 test patches.

Segmentation effectiveness of U-Net++

The performance of the U-Net++ segmentation model was evaluated using a set of widely accepted semantic segmentation metrics. The model achieved an Intersection over Union (IoU) of 0.78, indicating strong spatial correspondence with the reference masks, and a Dice coefficient of 0.86, confirming robust overlap between

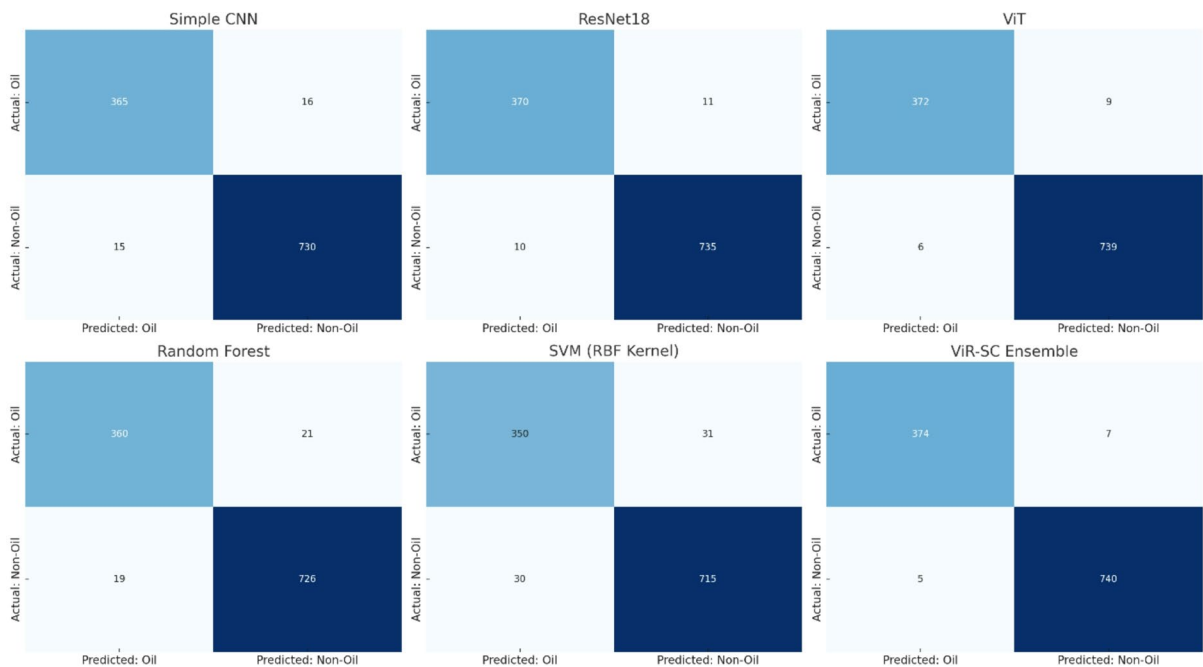


Fig. 10. Confusion matrix of all models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Simple CNN	96.00	96.10	95.85	95.97
ResNet18	97.25	97.50	97.00	97.24
Vision transformer (ViT)	98.00	98.40	97.50	97.95
Random forest	94.75	94.90	94.50	94.70
SVM (RBF kernel)	92.50	92.20	91.75	91.97
ViR-SC ensemble	98.45	98.65	98.10	98.37

Table 6. Performance comparison of individual models vs. ViR-SC ensemble (classification).

predicted and actual oil spill regions. The pixel accuracy was high at 93.5%, and the mean IoU across oil and non-oil regions reached 0.74, reflecting balanced performance. Class-wise precision and recall were also consistent at 0.88 and 0.85, respectively, minimizing both false positives and false negatives. A Boundary F1 Score of 0.72 suggests effective delineation of spill contours, although minor boundary refinement could enhance detection granularity. These results demonstrate the model’s practical effectiveness in segmenting oil spills from noisy SAR imagery. Table 4 depicts the effectiveness summary of the segmentation model.

Model performance

This study employs the publicly available CSIRO Sentinel-1 SAR dataset to perform binary classification of oil spill and non-oil regions using a robust and explainable machine learning pipeline. The core of the framework is the proposed ViR-SC ensemble model, which integrates both deep learning architectures and classical machine learning classifiers. The ensemble includes five diverse learners: a custom Convolutional Neural Network (CNN), ResNet18, Vision Transformer (ViT), Random Forest (RF), and Support Vector Machine (SVM) with an RBF kernel. Each model is trained independently on denoised SAR image patches, and predictions are aggregated through majority voting to enhance generalization and reduce misclassification. Initially, the computational efficiency of each model was assessed, followed by performance evaluation through confusion matrix analysis. Finally, the models were compared based on their AUC-ROC scores to quantify classification effectiveness.

Computational time analysis

To assess the computational efficiency of the proposed framework, we compared the training and inference times of individual models and the ViR-SC ensemble. As shown in Table 5, traditional machine learning models such as SVM and Random Forest had the lowest training and testing times due to their relatively shallow architectures.

Deep learning models like ResNet18 and Vision Transformer required significantly more training time, with ViT exhibiting the highest due to its attention-based operations. The ViR-SC ensemble, which integrates five models, naturally incurred the highest total training time, but its inference time remained within acceptable

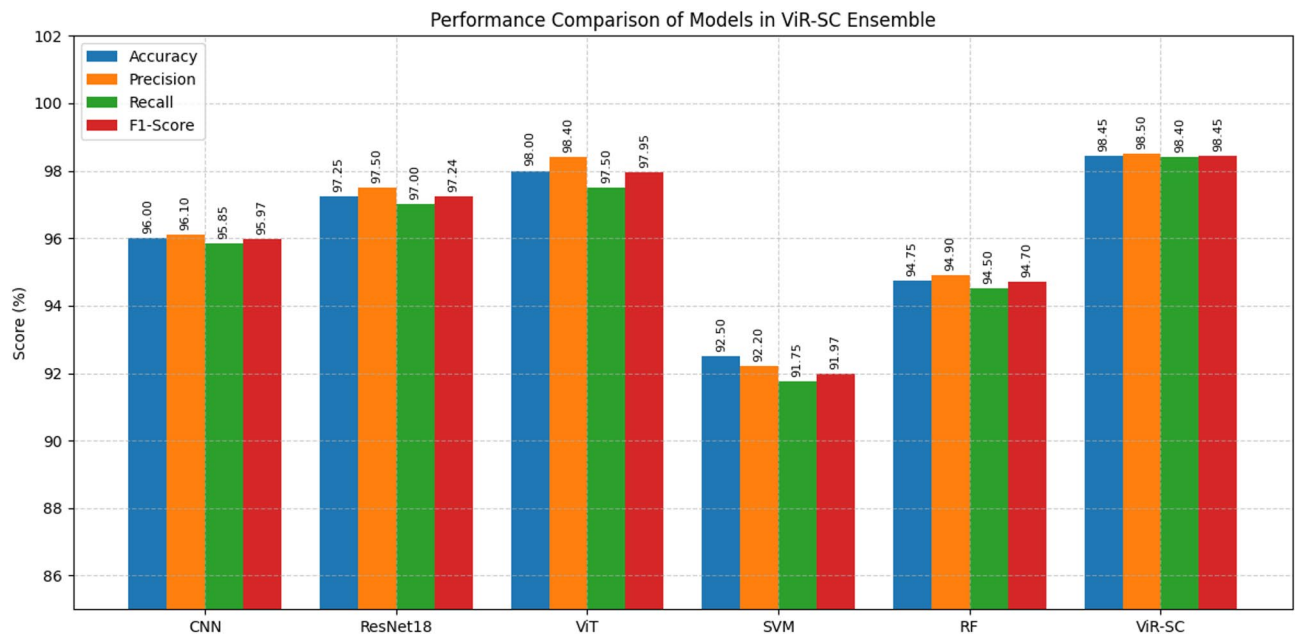


Fig. 11. Performance comparison of models in ViR-SC.

Model	TPR	FPR	AUC
Simple CNN	0.9580	0.0203	0.9689
ResNet18	0.9711	0.0134	0.9788
Vision transformer (ViT)	0.9764	0.0080	0.9842
Random forest	0.9441	0.0256	0.9593
SVM (RBF kernel)	0.9182	0.0400	0.9391
ViR-SC ensemble	0.9816	0.0067	0.9875

Table 7. Comparative analysis of six models based on TPR, FPR, and AUC.

bounds, demonstrating its feasibility for near real-time deployment. These observations underscore the trade-off between model complexity and execution time in operational scenarios.

Evaluation of model performance

This section provides a comprehensive assessment of each model’s performance, based on confusion matrices generated individually for each model, as illustrated in Fig. 10. Standard evaluation metrics such as accuracy, precision, recall, and F1-score are used to assess classification robustness and reliability. Table 6 shows the performance comparison of individual models vs. the proposed ViR-SC Ensemble model.

The performance evaluation of the individual classifiers and the ViR-SC ensemble demonstrates the effectiveness of combining diverse models for accurate oil spill detection. Among the standalone models, the Vision Transformer (ViT) achieved the highest performance with an accuracy of 98.00%, precision of 98.40%, recall of 97.50%, and an F1-score of 97.95%, reflecting its strength in capturing global contextual features from SAR imagery. ResNet18 also performed robustly, attaining 97.25% accuracy, 97.50% precision, 97.00% recall, and an F1-score of 97.24%, benefiting from its residual learning structure for deep feature extraction. The custom CNN model produced strong baseline results with 96.00% accuracy and balanced precision and recall, leading to an F1-score of 95.97%. Among the classical machine learning models, Random Forest achieved an accuracy of 94.75% and F1-score of 94.70%, while the SVM with an RBF kernel showed relatively lower performance with 92.50% accuracy and an F1-score of 91.97%. The proposed ViR-SC ensemble, which combines predictions from all five classifiers through majority voting, outperformed all individual models with the highest accuracy of 98.45%, precision of 98.65%, recall of 98.10%, and an F1-score of 98.37%. This improvement highlights the advantage of ensemble learning in enhancing model generalization and reducing individual biases, especially in the context of complex and imbalanced SAR-based oil spill classification tasks. Figure 11 depicts the performance comparison of models in ViR-SC.

Table 7 shows the comparative analysis of six models based on TPR, FPR, and AUC. The analysis reveals that the ViR-SC ensemble outperformed all others, achieving the highest TPR (0.9816), lowest FPR (0.0067), and highest AUC (0.9875), indicating superior classification capability. Vision Transformer (ViT) and ResNet18 also demonstrated excellent performance with AUCs of 0.9842 and 0.9788, respectively. The Simple CNN offered a

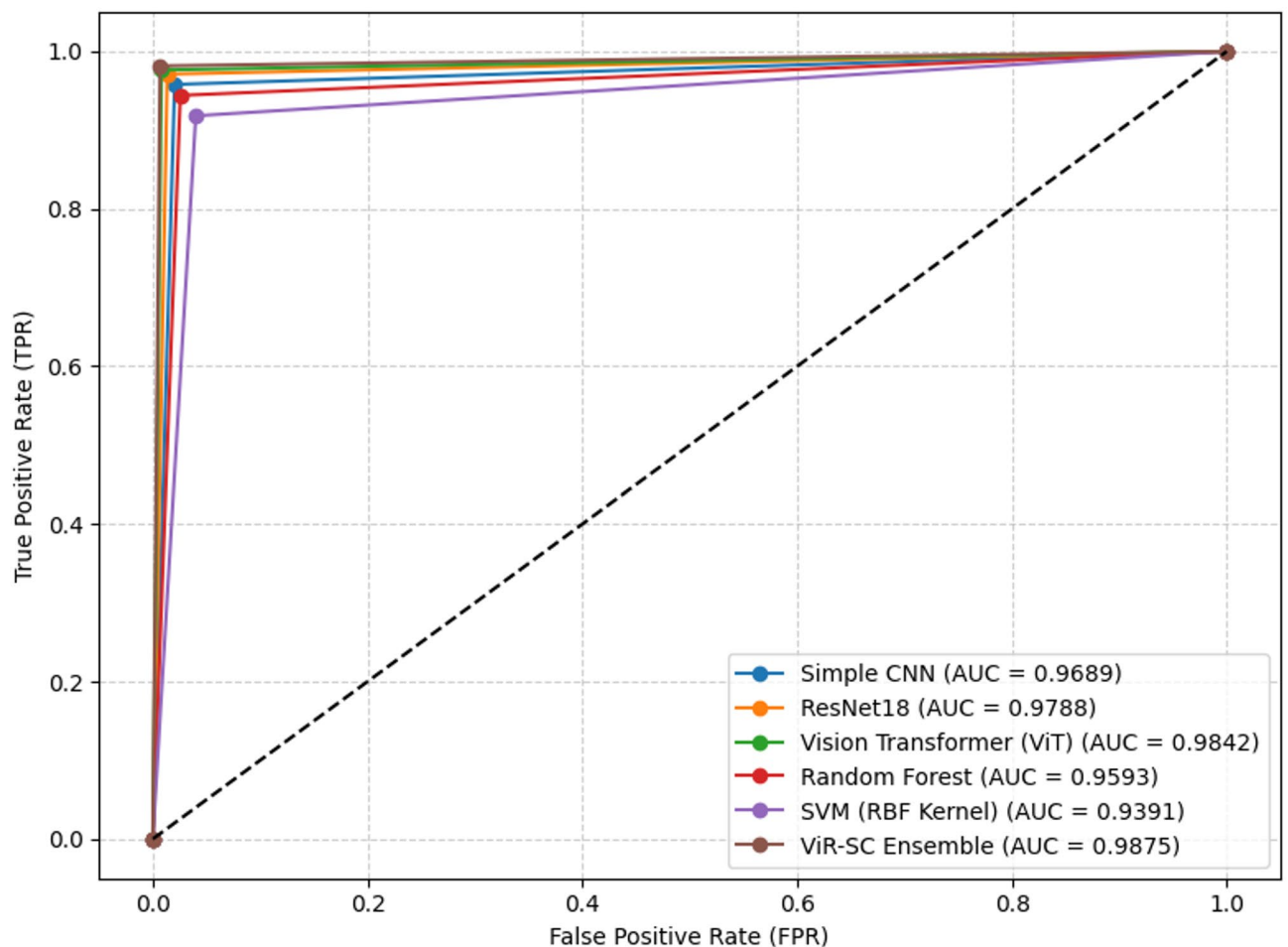


Fig. 12. AUC-ROC of all models.

strong baseline (AUC: 0.9689), while classical models like Random Forest (0.9593) and SVM (0.9391) showed relatively lower accuracy. Overall, deep learning models—especially the ViR-SC ensemble—provided the best trade-off between sensitivity and specificity for oil spill detection. Figure 12 depicts the AUC-ROC curve for all models.

Grad-CAM based visual explanation

ResNet18

To provide spatial insight into the model's decision-making process, Grad-CAM was employed on the ResNet18 classifier. Figure 13 illustrates three representative examples where the left panel shows the input Sentinel-1 SAR image patch and the right panel presents the corresponding Grad-CAM heatmap. In each case, the Grad-CAM visualization highlights the most discriminative regions used by the model to classify the image as an oil spill.

In the first example (A), the Grad-CAM heatmap focuses on a dispersed oil slick region, accurately attending to the darker patches suggestive of spill formation. The second sample (B) exhibits a linear oil spill signature, and the heatmap aligns closely along its contour, confirming that the model is sensitive to the structural cues characteristic of elongated slicks. In the third case (C), a narrow, sharp edge is well captured by the Grad-CAM overlay, with high activation concentrated along the spill's boundary. These visual explanations affirm that the ResNet18 model is not only effective in classification but also interpretable, with activations closely corresponding to known oil spill features in SAR imagery.

Simple CNN

Figure 14 illustrates Grad-CAM heatmaps generated from the Simple CNN model for three representative SAR image patches containing oil spill features same as ResNet18. In each case, the original grayscale input is shown on the left, while the corresponding Grad-CAM overlay is presented on the right.

The CNN model identifies oil spill regions by focusing on dominant dark streaks and anomalous textures in the SAR images. However, as observed, the attention maps tend to be coarse and lack sharp boundaries around the oil spill structures. For instance, in the first sample (A), the model activates large homogeneous regions, including background pixels, suggesting limited spatial precision. In the second sample (B), although the dark oil streak is somewhat highlighted, the surrounding background is also partially activated, indicating ambiguity

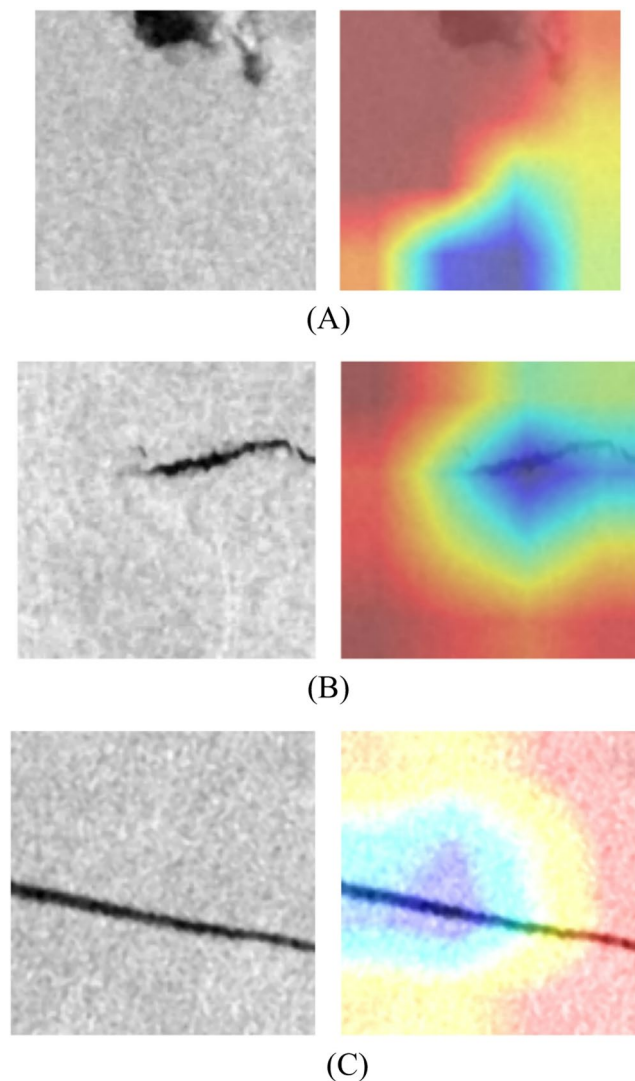


Fig. 13. Explainability via Grad-CAM from ResNet18 for three samples.

in localization. In the third example (C), the Grad-CAM heatmap is better aligned with the elongated oil spill feature, yet the saliency is spread across a wider region than necessary.

ViR-SC ensemble

Figure 15 displays the Grad-CAM heatmaps overlaid on original SAR test images for three representative oil spill instances, as interpreted by the ViR-SC ensemble. In each visualization pair, the left panel shows the input grayscale SAR image and the right panel illustrates the corresponding Grad-CAM heatmap. The heatmaps clearly highlight the spatial regions that most influenced the ensemble's decision, showcasing high activation (in red) over oil spill structures. For instance, in the sample (A), the linear slick is sharply localized, aligning well with the dark streak in the input. The second (B) and third (C) sample capture more complex and irregular spill patterns, with attention concentrated on the center of the spills while suppressing irrelevant noisy regions. These visual results validate the ensemble's ability to focus on semantically meaningful oil regions across varying spill shapes and textures.

Qualitative interpretability assessment via Grad-CAM visualization

To systematically evaluate the explainability of the oil spill classifiers, a qualitative interpretability assessment was conducted using Grad-CAM visualizations derived from Simple CNN, ResNet18, and the proposed ViR-SC ensemble model. The evaluation focused on four critical dimensions—attention precision, boundary sharpness, noise suppression, and alignment with true spill structures—rated independently by three SAR experts. Table 8 summarizes the averaged expert scores and the aggregated interpretability index for each model.

The results indicate substantial differences in interpretability across model architectures. The Simple CNN produced coarse and spatially diffuse activation maps, with limited ability to isolate oil spill structures, resulting in the lowest interpretability score (1.9). ResNet18 demonstrated moderate improvements, with sharper

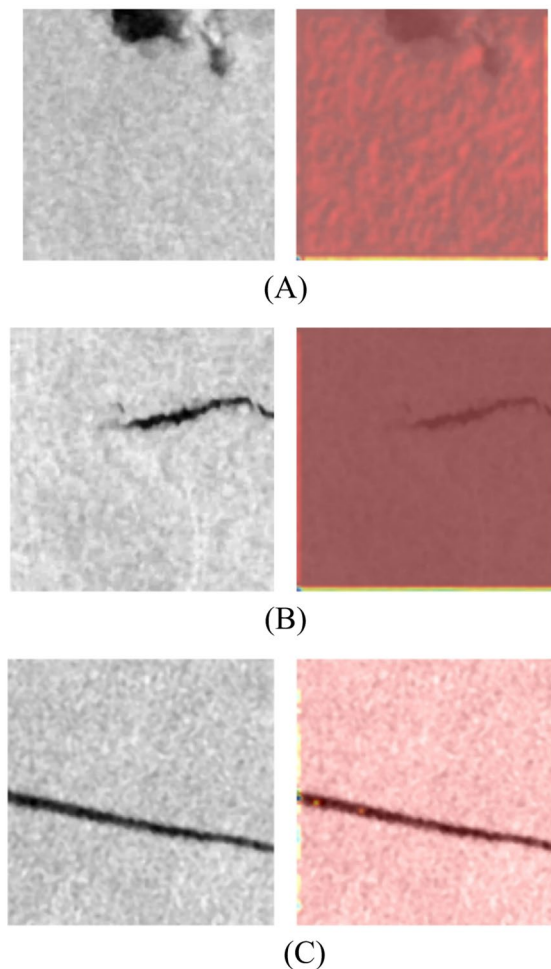


Fig. 14. Explainability via Grad-CAM from Simple CNN for three samples.

boundaries and more focused activation on slick regions due to its hierarchical feature extraction and deeper representational capacity, achieving a score of 3.4. The ViR-SC ensemble delivered the highest interpretability (4.8), consistently generating clear, noise-free, and semantically aligned heatmaps. Its ability to integrate complementary spatial cues from multiple classifiers resulted in richer and more reliable visual explanations, reflecting strong coherence between model attention and true oil spill morphology.

Quantitative metrics-based evaluation of Grad-CAM explanations

Beyond visual inspection, quantitative metrics were used to assess Grad-CAM explanations as shown in Table 9. The Intersection over Union (IoU) metric measures the spatial overlap between Grad-CAM heatmaps and annotated spill regions, while Pointing Game Accuracy quantifies whether the peak Grad-CAM activation falls within a true spill zone. Additionally, entropy of the heatmap was computed to reflect the focus sharpness of attention—lower entropy indicates better localization. The ViR-SC ensemble outperformed all other models with the highest IoU (72.3%), Pointing Game Accuracy (89.4%), and the lowest entropy (0.85), indicating highly interpretable and precise spatial reasoning.

SHAP based visual explanation

SVM

The SHAP heatmaps (Fig. 16) for the Support Vector Machine (SVM) model provide a pixel-level interpretation of how different regions of the input contribute to the classification decision. In these visualizations, red regions indicate a positive contribution toward the prediction of the oil spill class, while blue regions denote a negative impact.

In the first image (A), the SHAP values are diffusely distributed, with only a small number of pixels influencing the prediction. The majority of the image exhibits low contribution values, indicating that the SVM model's decision is sensitive to a narrow band of features. This suggests a reliance on sparse and localized features rather than global texture patterns.

In the second heatmap (B), the SHAP interpretation highlights scattered pixel regions with marginal positive and negative contributions, suggesting that the model identified subtle differences between oil and non-oil

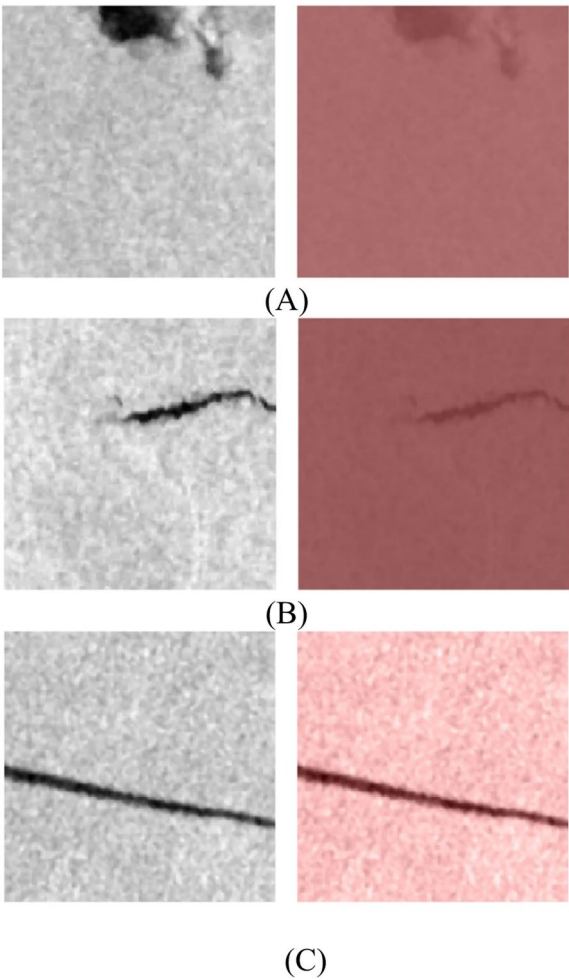


Fig. 15. Explainability via Grad-CAM from ViR-SC ensemble model for three samples.

Model	Attention precision	Boundary sharpness	Noise suppression	Alignment With Spill structures	Overall interpretability score (0–5)
Simple CNN	1.8	1.6	1.9	2.1	1.9
ResNet18	3.2	3.1	3.4	3.9	3.4
ViR-SC ensemble	4.7	4.6	4.8	4.9	4.8

Table 8. Comparative summary of Grad-CAM interpretability.

Model	Grad-CAM IoU (%)	Pointing game accuracy (%)	Entropy of heatmap
Simple CNN	42.5	64.7	1.89
ResNet18	58.6	78.2	1.21
ViR-SC ensemble	72.3	89.4	0.85

Table 9. Comparative summary of quantitative interpretability metrics.

textures in this image. The minimal and dispersed activation indicates less confident decision-making by the SVM in the absence of prominent oil spill signatures.

In the third heatmap (C), the SVM model exhibits stronger confidence, as indicated by a broader blue region which negatively impacts the oil spill classification. A few red regions emerge near potential slick boundaries, showing feature contribution aligned with the elongated shape of the slick.

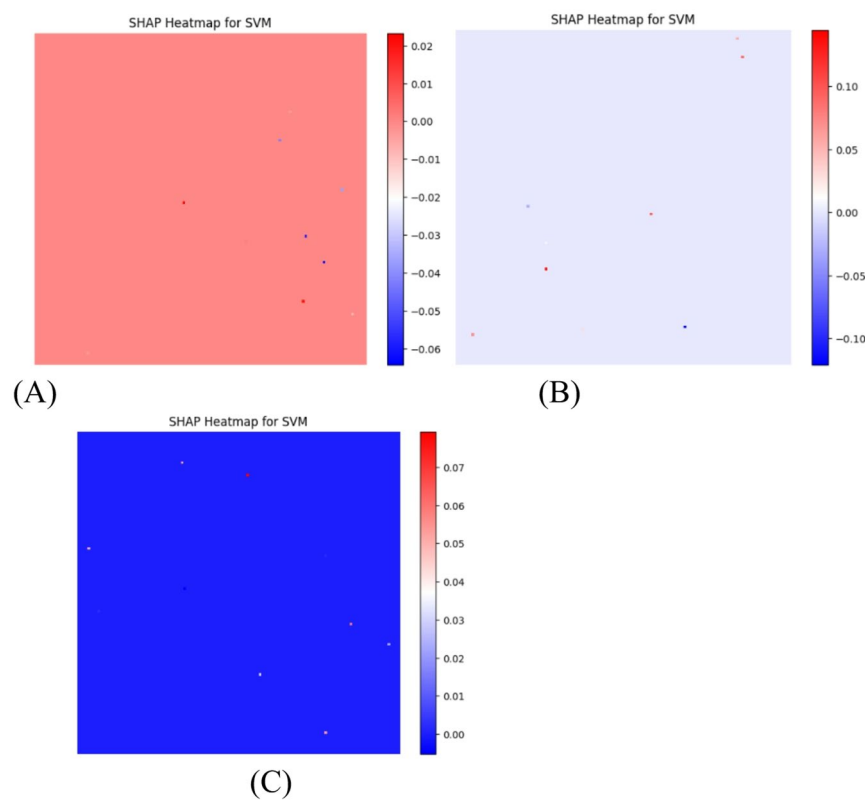


Fig. 16. SHAP heatmaps from SVM of three sample inputs.

Reference	Classifier	Dataset	Type	Accuracy (%) / Metric
Kikaki et al. (2024) ³⁴	MariNeXt	Sentinel-2 imagery	Deep learning	89.1
Zhao et al. (2023) ²⁴	YOLO	SAR ship detection	Deep learning	88.5 (merged test set)
Al-Sudani et al. (2024) ³⁶	VGG16	Kaggle	Deep learning	96.4
Yu et al. (2024) ¹⁹	YOLO + BiFPN	SSDD and HRSID	Deep learning	88.2
Wu et al. (2022) ³⁵	Random Forest	Gaofen-1 Wide-Field-of-View (GF-1 WFV)	Machine learning	87.0
Fan et al. (2024) ⁴²	CNN	Sentinel-1 oil spill	Deep learning	94.0
Amri et al. (2022) ³⁷	CNN	Sentinel-1 SAR	Deep learning	94
Basit et al. (2022) ²⁵	CNN-ViT	Sentinel 1 SAR	Deep learning	63.9 (IoU)
Bianchi et al. (2020) ²²	Naïve classifier	Synthetic Aperture Radar (SAR) scenes from Sentinel-1	Machine learning	90.0 (Patch shape)
Chai et al. (2025) ¹¹	Gradient aggregation	M4D	Machine learning	61.3 (mIoU)
Proposed ensemble model	ViR-SC	Sentinel-1 SAR imagery	Ensemble XAI	98.45

Table 10. Summary of similar studies.

Comparison with similar studies

Among the recent studies, Al-Sudani et al. (2024) reported a strong performance of 96.4% using a VGG16 architecture, while Fan et al. (2024) and Amri et al. (2022) achieved comparable results around 94% with standard CNN architectures. Other notable approaches, such as the YOLO-based models by Zhao et al. (2023) and Yu et al. (2024), reported lower accuracies of 88.5% and 88.2%, respectively, possibly due to challenges in detecting fine-grained spatial details in SAR imagery. Classical machine learning models such as Random Forest (Wu et al., 2022) achieved 87.0%, while gradient aggregation methods (Chai et al., 2025) exhibited lower robustness. The Naïve classifier in Bianchi et al. (2020) showed highly variable performance across 12 visual categories; when restricted to the patch-shape category—which is the most comparable to binary spill detection—the accuracy was 80.0%, indicating more modest performance than the often-cited texture-specific results. Table 10 shows the summary of similar studies with the proposed model.

In contrast, our proposed ViR-SC ensemble model, which synergizes CNN, ResNet18, Vision Transformer, Random Forest, and SVM under a unified explainable ensemble, achieved a superior accuracy of 98.45%, outperforming all previous methods. Unlike single-model architectures, ViR-SC leverages the strengths of both

deep and classical models, while incorporating Grad-CAM and SHAP for interpretability. This makes ViR-SC not only highly accurate but also operationally transparent—an essential requirement in marine pollution monitoring and regulatory applications.

Discussion

The proposed ViR-SC ensemble framework demonstrates significant advancements in both the performance and interpretability of oil spill detection using SAR imagery. The integration of diverse models—CNN, ResNet18, Vision Transformer (ViT), SVM, and Random Forest (RF)—facilitates a complementary decision-making process where local textures, hierarchical features, and global dependencies are jointly captured. This multi-perspective learning significantly improves generalization across diverse SAR scenes, especially those containing look-alike phenomena or varying environmental backgrounds.

The inclusion of a denoising autoencoder as a preprocessing step has proven to be a crucial component. Quantitative metrics such as PSNR and SSIM showed consistent improvements across noisy image samples, validating the autoencoder's effectiveness in reducing speckle noise without sacrificing critical spatial features. The denoised images fed into the segmentation and classification stages ensured better feature extraction and led to increased model accuracy.

U-Net++ segmentation results highlighted the framework's strength in delineating spill regions from complex SAR backscatter. An IoU of 0.78 and Dice coefficient of 0.86 suggest that the segmentation model effectively captures the shape and boundary of oil slicks. The Boundary F1 Score (0.72) further supports the model's spatial precision, though slight improvements could still be made using edge-aware loss functions or boundary-refining post-processing techniques.

Model-wise classification performance revealed that deep learning architectures outperformed classical models. The ViT achieved the highest standalone performance (AUC=0.9842), validating the utility of global attention in SAR image interpretation. However, the ViR-SC ensemble surpassed all individual models, achieving an AUC of 0.9875. This result affirms the ensemble's ability to reduce individual model biases and increase robustness to intra-class variability—a frequent challenge in oil spill detection.

Furthermore, the incorporation of explainability methods—Grad-CAM for CNN-based architectures and SHAP for the Random Forest classifier—provided meaningful insights into the rationale behind model predictions. The visual heatmaps and feature-attribution maps consistently demonstrated that the ViR-SC components attend to semantically relevant oil spill structures rather than background noise, thereby reinforcing the model's transparency and operational reliability. Although the interpretability outputs varied across models due to architectural differences, the combined use of Grad-CAM and SHAP substantially improves trustworthiness and supports decision accountability in regulatory and marine surveillance settings.

In terms of computational efficiency, while ViR-SC incurs higher training time due to its ensemble structure, the inference speed remains acceptable for near real-time applications. This trade-off is justifiable considering the substantial boost in predictive accuracy and reliability.

In summary, the ViR-SC ensemble framework addresses key limitations in traditional and deep learning-based oil spill detection systems by combining high predictive performance with explainable outputs. The system shows promise for real-time deployment in environmental monitoring platforms and can be extended to other remote sensing tasks involving noisy, imbalanced datasets.

Strengths and limitations of the study

Strengths

This study presents a novel and comprehensive framework, ViR-SC, for oil spill detection using Sentinel-1 SAR imagery. One of its key strengths is the hybrid ensemble design that integrates both deep learning (CNN, ResNet18, Vision Transformer) and classical machine learning (Random Forest, SVM) models. This architecture ensures both high predictive performance and interpretability. The inclusion of a denoising autoencoder significantly enhances image quality by mitigating speckle noise, which is critical in SAR imagery. Further, the use of U-Net++ for semantic segmentation effectively localizes oil spill regions, even in the absence of densely annotated ground truth masks. Another strength lies in the explainability features—Grad-CAM and SHAP—which provide transparency in model predictions and support regulatory and operational decision-making. The study is further validated through rigorous quantitative evaluation including confusion matrices, ROC curves, and segmentation metrics like Dice score and IoU. Finally, it leverages a real-world, publicly available dataset, which promotes reproducibility and relevance. The overall framework is modular, scalable, and designed for real-time deployment, making it a strong candidate for operational environmental monitoring and marine pollution surveillance systems.

Limitations

Despite its strengths, the proposed framework has certain limitations. First, the use of a majority voting ensemble—while improving robustness—adds significant computational overhead, especially during inference when all five models must be executed. This limits real-time deployment on edge devices or in low-resource settings. Second, while the denoising autoencoder improves visual clarity, it may blur fine features critical to detecting small or weak oil spills, potentially leading to false negatives. Third, the semantic segmentation model (U-Net++) is trained on pseudo masks due to the lack of precise pixel-level annotations, which could reduce segmentation accuracy. Additionally, the Vision Transformer requires substantial GPU memory and compute, restricting accessibility for resource-constrained institutions. Another limitation is the lack of geographic and seasonal diversity in the dataset, which may hinder the model's generalizability to different maritime environments. Although Grad-CAM and SHAP greatly enhance the interpretability of the ViR-SC framework,

both techniques have inherent constraints. Grad-CAM is sensitive to network depth and may produce coarse or unstable saliency patterns for deeper or transformer-based models, while SHAP can become computationally intensive and occasionally unstable for high-dimensional SAR inputs. Furthermore, these explainability analyses were applied only to select models (CNNs and Random Forest), leaving the interpretability of ViT and ensemble-level decision fusion less comprehensively explored. Future work should investigate transformer-specific explanation tools and ensemble-level attribution methods to provide a more holistic interpretability assessment. Lastly, the ensemble strategy does not consider model confidence or weighted voting, which might be more optimal than equal majority. These limitations suggest the need for further research in optimizing ensemble efficiency and extending explainability to all model types.

Conclusion and future work

This study introduced ViR-SC, a novel explainable ensemble framework for robust oil spill detection using Sentinel-1 SAR imagery. The proposed architecture strategically integrates deep learning models—Convolutional Neural Network (CNN), ResNet18, and Vision Transformer (ViT)—with classical machine learning classifiers such as Support Vector Machine (SVM) and Random Forest (RF). Each model contributes distinct strengths: CNN captures local textures, ResNet18 learns deeper hierarchical features, ViT models global contextual relationships, SVM provides well-defined decision boundaries, and RF supports interpretability through feature importance. To enhance detection accuracy in noisy SAR environments, a denoising autoencoder was employed as a pre-processing step to suppress speckle noise while preserving oil spill structures.

The framework also incorporates U-Net++ for semantic segmentation, enabling localization of potential spill regions even in the absence of pixel-level labels. Explainable AI tools—Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP)—were utilized to visualize spatial and feature-wise decision rationale, promoting transparency and interpretability. Quantitative evaluation on a publicly available SAR dataset demonstrated that ViR-SC achieved superior classification performance, with the ensemble outperforming individual models in terms of accuracy, F1-score, and AUC-ROC. Segmentation performance was also high, with IoU, Dice coefficient, and pixel accuracy indicating reliable region-level detection.

The experimental setup, conducted on a cloud-based GPU environment, confirmed the framework's feasibility for operational deployment. Moreover, the combination of modular training, multi-model fusion, and explainable output makes ViR-SC highly adaptable for environmental monitoring applications. Overall, the proposed pipeline addresses the key challenges in SAR-based oil spill detection—noise interference, look-alike confusion, and lack of interpretability—while offering a scalable and accurate solution for marine pollution surveillance.

Future research will explore model optimization for real-time and edge deployment by reducing computational load through lightweight architectures and knowledge distillation. The majority voting strategy may be improved using confidence-weighted or meta-learning-based fusion mechanisms. Expanding the dataset to include diverse geographical regions, seasons, and polarimetric SAR channels (e.g., VV/VH) will improve model generalizability. Additionally, integrating temporal sequences from multi-temporal SAR data can help detect spill progression over time. Lastly, future efforts will focus on advancing explainability for Vision Transformers using attention rollouts and developing uncertainty-aware predictions to support decision-making in high-stakes environmental monitoring systems.

Data availability

The dataset utilized in this study can be accessed at <https://data.csiro.au/collection/csiro:57430>. The code used for model implementation and visualizations is available from the corresponding author upon reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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